BE-204 · BTech First Year · UIT, Barkatullah University, Bhopal

Basic Civil Engineering & Engineering Mechanics — Short Notes

Unit-wise complete syllabus coverage · Hinglish notes · Prepared by Ajeet Soni

UNIT I Statics & Friction

1. Coplanar Forces & Resultant

Force ek vector quantity hai (magnitude, direction, point of application, line of action). Jab saari forces ek hi plane mein ho to unhe **coplanar forces** kehte hain.

Type	Meaning	Equilibrium conditions
Collinear	Sab forces ek hi line of action par	$\Sigma F = 0$
Concurrent	Sab forces ek point se pass hoti hain	$\Sigma F_x = 0$, $\Sigma F_y = 0$
Parallel (like / unlike)	Lines of action parallel	$\Sigma F_y = 0$, $\Sigma M = 0$
Non-concurrent, non-parallel (general)	Na ek point, na parallel	$\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma M = 0$

FORMULA Resultant (resolution method):

$$R = \sqrt{[(\Sigma F_x)^2 + (\Sigma F_y)^2]}, \quad \tan\alpha = \Sigma F_y / \Sigma F_x$$

Two forces P, Q at angle θ (parallelogram law): $R = \sqrt{(P^2 + Q^2 + 2PQ\cos\theta)}$, $\tan\alpha = Q\sin\theta / (P + Q\cos\theta)$.
Lami's theorem (3 concurrent forces in equilibrium): $P/\sin\alpha = Q/\sin\beta = R/\sin\gamma$ (α = Q aur R ke beech ka angle, yaani P ke opposite wala angle; isi tarah β, γ).

2. Free Body Diagram (FBD)

FBD mein body ko surroundings se **alag (isolate)** karke uspar lagne wali saari forces dikhate hain: weight W (downward, CG par), applied loads, support reactions, friction, tension. Steps: (1) body isolate karo, (2) weight lagao, (3) har contact/support ko uske reaction se replace karo, (4) unknown directions assume karo (answer negative aaye to direction ulti).

Supports in FBD: smooth surface / roller → 1 reaction (surface ke normal); hinge / pin → 2 (R_x, R_y); fixed → 3 (R_x, R_y, M); string / cable → tension (body se door, pull).

3. Moment & Varignon's Theorem

Moment = Force × perpendicular distance (M = F·d), unit N·m. Clockwise / anticlockwise sign convention fix rakho. **Couple** = do equal, opposite, parallel forces → sirf rotation, moment = F × arm, kisi bhi point ke about same.

VARIGNON'S THEOREM Kisi point ke about forces ke system ke **resultant ka moment** = usi point ke about saari **individual forces ke moments ka algebraic sum**. $R \cdot d = \Sigma F \cdot x$. Use: resultant ki **line of action** (position) nikalne mein: $d = \Sigma M / R$.

Equilibrium conditions: Body equilibrium mein hai jab resultant force bhi zero aur resultant moment bhi zero: **$\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma M = 0$** (coplanar mein max 3 unknowns solve ho sakte hain). Two-force member: dono forces equal, opposite, collinear. Three-force member: forces concurrent ya parallel honi chahiye.

4. Polygon Law of Forces & Funicular Polygon (Graphical)

- **Polygon law:** Agar kai concurrent forces ko magnitude + direction mein ek order mein (tip to tail) draw karein, to **closing side ulti order mein** resultant deti hai. Polygon close ho jaye → resultant zero → equilibrium (triangle law iska special case hai).
- **Bow's notation:** Forces ke beech ke spaces ko A, B, C… se naam do; force ko do letters (AB) se denote karo — graphical work easy.
- **Funicular (link/string) polygon:** Non-concurrent forces ke liye. Force polygon banao → ek **pole O** choose karo → rays (Oa, Ob…) draw karo → space diagram mein rays ke parallel links draw karo. Pehle aur last link ka intersection = resultant ki **line of action**.
- **Graphical equilibrium:** Force polygon close → no resultant force; funicular polygon close → no resultant couple. Dono close → body in equilibrium. Beam ke support reactions isi se nikalte hain.

5. Analysis of Perfect Truss

Truss = straight members ka framework jo ends par pin-joint hote hain; load sirf joints par; members sirf axial force (Tension T ya Compression C) lete hain (self-weight neglect).

FORMULA

$$\text{Perfect truss: } m = 2j - 3 \;\mid\; m < 2j - 3 \rightarrow \text{deficient} \;\mid\; m > 2j - 3 \rightarrow \text{redundant}$$

(m = members, j = joints)

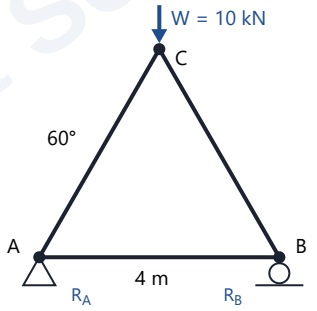
Method of Joints

1. Pehle poore truss ke support reactions (ΣM, ΣF_y, ΣF_x).
2. Aisa joint chuno jahan **≤ 2 unknown** members ho.
3. $\Sigma F_x = 0$, $\Sigma F_y = 0$ lagao. Force joint se door → Tension; joint ki taraf → Compression.
4. Joint-by-joint aage badho.

Method of Sections

1. Reactions nikalo.
2. Section line se truss ko kaato — **max 3 unknown** members cut hon (sab ek joint par na milte ho).
3. Ek part ka FBD lo; ΣM about intersection of 2 unknowns → teesra directly.

Use: beech ke kuch members ki force jaldi chahiye ho to sections best.



Equilateral truss, load W at apex C

SOLVED Symmetry se $R_A = R_B = 5$ kN.
Joint A: ΣF_y: $F_{AC} \sin 60° = 5 \rightarrow$ **F_{AC} = 5.77 kN (C)**.
ΣF_x: $F_{AB} = F_{AC} \cos 60° =$ **2.89 kN (T)**. Symmetry: F_{BC} = 5.77 kN (C). Check m = 3, j = 3: 2(3)−3 = 3 → perfect.

YAAD RAKHO Joint par sirf 2 non-collinear members aur koi load nahi → dono members ki force zero (**zero-force members**). 3 members, 2 collinear, no load → teesra zero.

6. Friction

Friction = contact surfaces ke beech relative motion ko oppose karne wali force. Static (limiting) > Dynamic (kinetic). **Laws of Coulomb friction:** F ∝ N; F direction motion ke opposite; F contact area par depend nahi karta; kinetic friction velocity se (lagbhag) independent.

FORMULA

$$F = \mu N, \quad \mu = \tan\varphi \;(\varphi = \text{angle of friction}), \quad \text{angle of repose } \alpha = \varphi$$

Resultant reaction S = $\sqrt{(N^2 + F^2)}$, N ke saath angle φ banata hai.

(a) Friction on Inclined Plane (weight W, angle θ)

Effort P direction	Moving UP the plane	Moving DOWN the plane
Parallel to plane	P = W(sin θ + μ cos θ)	P = W(μ cos θ − sin θ) (θ < φ)
Horizontal	P = W tan(θ + φ)	P = W tan(φ − θ)
At angle α to plane	P = W sin(θ + φ) / cos(α − φ)	P = W sin(φ − θ) / cos(α + φ)

Least effort to drag up: jab α = φ, tab P_{min} = W sin(θ + φ). **Example:** W = 500 N, θ = 30°, μ = 0.2, P parallel up: P = 500(0.5 + 0.2×0.866) = **336.6 N**.

(b) Screw Jack

Screw jack ek inclined plane hai jo cylinder par lipta hua hai. Thread ko unroll karo: base = πd, height = pitch p. Load W ko effort P (mean radius par) ghuma ke uthate hain.

FORMULA

$$\tan\alpha = p / (\pi d); \quad P = W\tan(\alpha + \varphi); \quad \text{Handle effort } P_1 = P\cdot(d/2)/L$$

$$\eta = \tan\alpha / \tan(\alpha + \varphi), \quad \eta_{max} = (1 - \sin\varphi)/(1 + \sin\varphi); \quad \text{Lowering torque} = W\tan(\varphi - \alpha)\cdot d/2$$

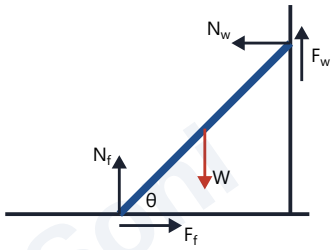
Multi-start thread: lead = n × pitch, tan α = lead/(πd). **Self-locking** (load khud neeche nahi aata) jab φ > α; tab η < 50%.

Example: d = 50 mm, p = 10 mm, μ = 0.1, W = 20 kN, handle L = 500 mm. tan α = 10/(π×50) = 0.0637 (α = 3.64°), φ = 5.71°. P = 20000 × tan 9.35° = **3294 N**; handle effort = 3294×25/500 = **164.7 N**; η = 0.0637/0.1647 = **38.6%** → self-locking.

(c) Ladder Friction

Ladder (length l, weight W at middle) floor aur wall par tika hai. Ladder ka foot wall se **door** slip karta hai, isliye floor friction F_f wall ki **taraf**; top neeche slip karta hai, isliye wall friction F_w **upar**.

Equations: $\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma M_{foot} = 0$, with $F_f = \mu_f N_f$, $F_w = \mu_w N_w$ (limiting).



Ladder FBD (foot slips away from wall)

Example: Smooth wall, floor $\mu = 0.25$, ladder ka angle θ (floor se) on point of slipping? $N_f = W$, $N_w = F_f = 0.25W$. ΣM about foot: $N_w \cdot l \sin\theta = W \cdot (l/2)\cos\theta \rightarrow \tan\theta = 1/(2\mu) = 2 \rightarrow$ **θ = 63.4°**.

(d) Wedge Friction

Wedge ek chhota tapered block hai jo heavy load ko thoda uthane / adjust karne ke liye use hota hai. Effort P wedge par lagta hai; har contact surface par friction motion ke opposite. Method: wedge aur block ka **alag-alag FBD** banao; har surface par reaction R normal se φ angle par (Lami's theorem ya ΣF_x, ΣF_y). Pehle block ki equilibrium se unknown reaction, phir wedge se P.

EXAM TIP Friction numericals mein sabse common galti friction ki direction hai — pehle socho body kis taraf move "karna chahti" hai, friction uske ulta lagao. Ladder, block-on-incline aur screw jack (η, self-locking) har saal aate hain.

UNIT II Centroid, Moment of Inertia & Power Transmission

1. Centre of Gravity & Centroid of Plane Areas

Centre of gravity (CG) = woh point jahan body ka poora weight act karta hua maana jaata hai. Plane area (no weight) ke liye isi point ko **centroid** kehte hain. Symmetric figure mein centroid axis of symmetry par hota hai.

FORMULA Composite area:

$$\bar{x} = \Sigma A_i x_i / \Sigma A_i, \quad \bar{y} = \Sigma A_i y_i / \Sigma A_i \quad \text{(hole/cut-out ka area negative lo)}$$

Shape	Area	Centroid	I about centroidal axis (I_{xx})	I about base/diameter
Rectangle b × d	bd	d/2 from base	$bd^3/12$	$bd^3/3$
Triangle (base b, height h)	bh/2	h/3 from base	$bh^3/36$	$bh^3/12$ (apex se parallel axis: $bh^3/4$)
Circle (dia d)	$\pi d^2/4$	centre	$\pi d^4/64$	$I_{polar} = \pi d^4/32$
Semicircle (radius r)	$\pi r^2/2$	4r/(3π) from diameter	$0.11\ r^4$	$\pi r^4/8 = \pi d^4/128$
Quarter circle	$\pi r^2/4$	4r/(3π) from both edges	$0.055\ r^4$	$\pi r^4/16$

2. Moment of Inertia (Area) & Theorems

Area MOI (second moment of area) $I = \Sigma(dA)\cdot y^2 = \int y^2 dA$, unit mm^4/m^4. Bending strength ka measure. **Radius of gyration** k = √(I/A).

PARALLEL AXIS THEOREM

$$I_{AB} = I_G + A\,h^2$$

h = centroidal axis aur parallel axis AB ke beech distance. (Centroidal I hamesha minimum.)

PERPENDICULAR AXIS THEOREM

$$I_{zz} = I_{xx} + I_{yy}$$

Plane lamina ke liye; z-axis plane ke perpendicular, x, y ke intersection se. Example: circle $I_{polar} = 2 \times \pi d^4/64$.

SOLVED (T-SECTION) Flange 100×20 mm (top), web 20×80 mm. Base se: $A_1 = 2000$, $y_1 = 90$; $A_2 = 1600$, $y_2 = 40$. $\bar{y}$ = (2000×90 + 1600×40)/3600 = **67.78 mm** (base se).
$I_{xx} = [100\times20^3/12 + 2000\times22.22^2] + [20\times80^3/12 + 1600\times27.78^2] = (66{,}667 + 9{,}87{,}654) + (8{,}53{,}333 + 12{,}34{,}568) =$ **3.14×10^6 mm⁴**.

3. Mass Moment of Inertia

Rotation ke against resistance (rotational inertia): $I = \Sigma m\, r^2 = \int r^2 dm = Mk^2$, unit kg·m^2. Parallel axis theorem yahan bhi: $I = I_G + Mh^2$. $KE_{rot} = \frac{1}{2} I\omega^2$, Torque $T = I\alpha$.

Body	Axis	I	Body	Axis	I
Thin rod (L)	centre, ⊥ rod	$ML^2/12$	Thin rod	end, ⊥ rod	$ML^2/3$
Thin ring (R)	polar (⊥ plane)	MR^2	Solid disc / cylinder	own axis	$MR^2/2$
Solid sphere	diameter	$2MR^2/5$	Hollow sphere (thin)	diameter	$2MR^2/3$
Rect. plate b×d	centroidal, ∥ b	$Md^2/12$	Solid cylinder	centroidal, ⊥ axis	$M(3R^2 + L^2)/12$

4. Power Transmission — Belts & Ropes

Belts/ropes power ko ek shaft se doosre shaft tak **friction** ke through pahunchate hain. Types: flat belt (moderate distance, ~10 m tak), V-belt (chhoti distance, zyada grip, slip kam), circular rope (lambi distance, jaise cranes/mills). Drives: **open belt** (same direction rotation) aur **cross belt** (opposite direction).

FORMULA

$$\text{Velocity ratio: } N_2/N_1 = d_1/d_2 \quad [\text{thickness t ke saath: } (d_1+t)/(d_2+t);\ \text{slip s\% ke saath} \times (1 - s/100)]$$

$$\text{Length: Open } L = \pi(r_1+r_2) + 2x + (r_1-r_2)^2/x; \quad \text{Cross } L = \pi(r_1+r_2) + 2x + (r_1+r_2)^2/x$$

$$\text{Angle of contact: Open } \theta = 180° - 2\alpha,\ \sin\alpha = (r_1-r_2)/x; \quad \text{Cross } \theta = 180° + 2\alpha,\ \sin\alpha = (r_1+r_2)/x$$

$$\text{Ratio of tensions (flat): } T_1/T_2 = e^{\mu\theta}; \quad \text{V-belt/rope (groove angle } 2\beta\text{): } T_1/T_2 = e^{\mu\theta\ \mathrm{cosec}\,\beta}$$

$$\text{Power } P = (T_1 - T_2)\cdot v \text{ watt}, \quad v = \pi dN/60 \text{ m/s}$$

$$\text{Centrifugal tension } T_c = mv^2; \quad \text{Max tension } T = T_1 + T_c = \sigma\cdot b\cdot t; \quad \text{Max power when } T_c = T/3,\ v = \sqrt{(T/3m)}$$

θ hamesha **radian** mein, aur smaller pulley ka angle of contact lo (open belt). T_1 = tight side, T_2 = slack side, m = mass per metre length. Initial tension $T_0 = (T_1 + T_2)/2$ (+ T_c if considered).

EXAMPLE Flat belt, μ = 0.3, θ = 180° = π rad, T_1 = 1000 N, v = 10 m/s. $T_1/T_2 = e^{0.3\pi} = 2.566 \rightarrow T_2 = 389.7$ N. P = (1000 − 389.7)×10 = **6.10 kW**.

5. Gears & Gear Trains

Gears = toothed wheels; positive drive (no slip), exact velocity ratio, short centre distance, high power. **Module** m = d/T (mm), circular pitch $p_c = \pi d/T = \pi m$.

Gear type	Shafts	Use / remark
Spur	Parallel, straight teeth	Simple, noisy at high speed — lathe, gearbox
Helical	Parallel, inclined teeth	Smooth, quiet; axial thrust (double helical/herringbone se cancel)
Bevel	Intersecting (usually 90°)	Differential of automobile; mitre gears (equal size)
Worm & worm wheel	Non-intersecting, ⊥	Very high speed reduction, self-locking possible
Rack & pinion	Rotary ↔ linear	Lathe carriage, steering

FORMULA

$$\text{Two gears: } N_1/N_2 = T_2/T_1 = d_2/d_1 \quad (\text{external mesh} \rightarrow \text{opposite direction})$$

$$\text{Simple train: } N_1/N_n = T_n/T_1 \ (\text{idlers ratio change nahi karte, sirf direction; odd no. of idlers} \rightarrow \text{same direction})$$

$$\text{Compound train: } N_{last}/N_{first} = (\text{Product of teeth on drivers})/(\text{Product of teeth on driven}) = \text{Train value}$$

- **Simple gear train:** har shaft par ek gear. **Compound:** ek shaft par do gears (badi reduction compact space mein). **Reverted:** compound train jismein first aur last gear co-axial (clock, lathe back gear); condition $r_1 + r_2 = r_3 + r_4$. **Epicyclic:** ek gear ka axis doosre ke around ghoomta hai (differential, hoist) — tabular method se solve.
- **Example (compound):** A (20 T, driver, 1500 rpm) meshes B (60 T); C (25 T) B ke shaft par, meshes D (75 T). $N_D = 1500 \times (20\times25)/(60\times75)$ = **166.7 rpm**.

Belt vs Gear: belt mein slip → VR exact nahi, large centre distance, shock absorb, cheap; gear mein no slip → exact VR, small centre distance, high power, compact, par costly aur lubrication chahiye.

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UNIT III Shear Force & Bending Moment

1. Types of Beams & Loading

Beam = structural member jo apni length ke perpendicular (transverse) load leta hai aur bending se resist karta hai.

Types of beams: cantilever (ek end fixed, doosra free), **simply supported** (dono ends par hinge/roller), **overhanging** (ek/dono end support ke bahar nikle) — ye teeno **statically determinate** (3 equations se reactions); **fixed** (dono ends fixed), **propped cantilever** (cantilever + free end par prop), **continuous** (2 se zyada supports) — indeterminate.

Types of loading: point/concentrated load W (kN) ek point par; **UDL** w (kN/m) — total wL, centre par act; **UVL** (triangular/trapezoidal) — triangle total wL/2, bade end se L/3 par act; **applied couple/moment** M (kN·m).

DEFINITIONS **Shear Force (SF)** at a section = section ke ek side (left ya right) ki saari vertical forces ka algebraic sum. **Bending Moment (BM)** at a section = section ke ek side ki saari forces ke moments ka algebraic sum (section ke about). **Sign convention:** Left side upward force → +SF; **sagging** (concave upward, beam "smile") → +BM, **hogging** → −BM.

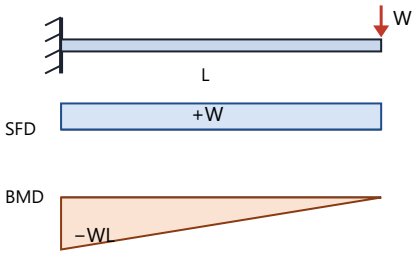
2. Relationship between Load, SF and BM

FORMULA

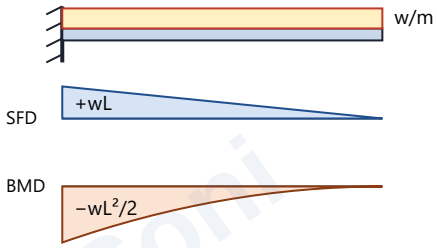
$$\frac{dF}{dx} = -w \quad \text{(load intensity)} \qquad | \qquad \frac{dM}{dx} = F \quad \text{(shear force)}$$

- BM **maximum** wahan hota hai jahan SF = 0 (ya sign change karta hai).
- No load portion → SF constant (horizontal line), BM linear (inclined line).
- UDL → SF linear, BM parabolic (2nd degree). UVL → SF parabolic, BM cubic.
- Point load par SF mein sudden jump (load ke barabar); applied couple par BM mein sudden jump.
- Do sections ke beech BM ka change = SF diagram ka area: $M_2 - M_1 = \int F\,dx$.

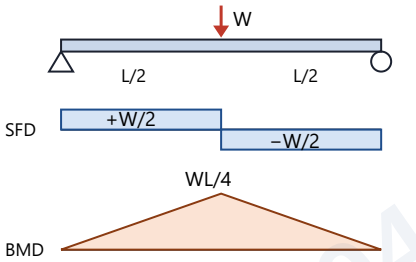
3. Standard SFD & BMD Results



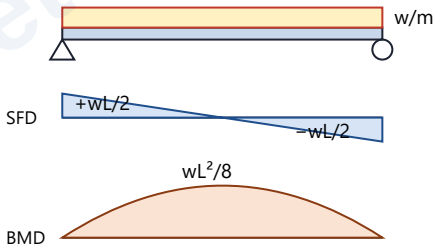
Cantilever, point load W at free end



Cantilever, UDL w over full span (SF linear, BM parabola)



Simply supported, central point load W



Simply supported, UDL w over full span

Beam & load	Reactions	SF_{max}	BM_{max} (location)
Cantilever, point load W at free end	R = W, M_{fix} = WL	W (constant)	−WL (fixed end)
Cantilever, UDL w full span	R = wL	wL (fixed end)	$-wL^2/2$ (fixed end)
Cantilever, UVL 0 (free) → w (fixed)	R = wL/2	wL/2	$-wL^2/6$ (fixed end)
SS, central point load W	W/2 each	W/2	WL/4 (centre)
SS, eccentric W at a from A (b from B)	R_A = Wb/L, R_B = Wa/L	larger reaction	Wab/L (under load)
SS, UDL w full span	wL/2 each	wL/2	$wL^2/8$ (centre)
SS, UVL 0 (A) → w (B)	R_A = wL/6, R_B = wL/3	wL/3	$wL^2/(9\sqrt{3})$ at x = L/√3 from A

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4. Overhanging Beam & Point of Contraflexure

POINT OF CONTRAFLEXURE Woh point jahan BM **sign change** karta hai (sagging ↔ hogging), yaani BM = 0 (free/simple end ke alawa). Ye overhanging (aur fixed/continuous) beams mein milta hai, cantilever ya simple SS beam mein nahi. Wahan curvature ulti hoti hai — RCC mein reinforcement yahan bend/cut karte hain.

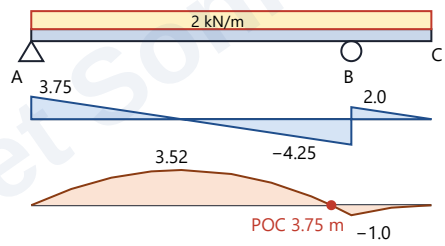
Example: Beam ABC, AB = 4 m (supports A, B), overhang BC = 1 m, UDL 2 kN/m over full 5 m.

Reactions: total load 10 kN at 2.5 m from A. ΣM_A: $R_B \times 4 = 10 \times 2.5$ → **R_B = 6.25 kN**, R_A = **3.75 kN**.

SF: A = +3.75; B (left) = 3.75 − 8 = −4.25; B (right) = −4.25 + 6.25 = +2.0; C = 0.

BM ($0 \le x \le 4$): $M = 3.75x - x^2$. SF = 0 at x = 1.875 m → **M_{max} = +3.52 kN·m**. $M_B = -2 \times 1^2/2$ = **−1 kN·m**.

Contraflexure: $3.75x - x^2 = 0$ → **x = 3.75 m** from A.



SFD (kN) and BMD (kN·m) of overhanging beam

EXAM TIP SFD/BMD steps: (1) reactions ($\Sigma M = 0$, $\Sigma F_y = 0$) aur check, (2) har load point par SF, (3) SF = 0 wala point dhoondo → M_{max}, (4) supports aur load points par BM, (5) curves ka shape (constant/linear/parabola) relationship se, (6) contraflexure M = 0 equation se. Values diagram par zaroor likho.

UNIT IV Building Materials & Construction

1. Stones

Geological class	Formation	Examples
Igneous	Molten magma ke cool hone se; hard, durable	Granite, basalt, trap
Sedimentary	Sediments ki layers jamne se; stratified	Sandstone, limestone, laterite
Metamorphic	Heat + pressure se purane rock ka badalna	Marble (limestone se), slate (shale se), quartzite

Physical classification: stratified, unstratified, foliated. Chemical: siliceous, argillaceous, calcareous.

Properties of good stone: high crushing strength (>100 N/mm^2 for building), water absorption < 5%, hardness, toughness, durability, fire resistance, uniform texture, easy dressing, specific gravity > 2.7.

Tests: crushing strength test, water absorption test, attrition (Deval) test (wear), hardness (Mohs scale / Dorry's), impact (toughness) test, acid test (weathering), freezing-thawing test, microscopic test.

Uses: granite — bridge piers, steps, monuments; marble — flooring, facing; slate — roofing, DPC; sandstone — walls; basalt/trap — road metal, aggregates; limestone — cement and lime manufacture.

2. Bricks

Composition of good brick earth: silica 50–60% (shape), alumina 20–30% (plasticity), lime 5–10% (silica ko melt hone mein help, shrinkage kam), iron oxide 5–6% (red colour, strength), magnesia < 1%. Harmful: excess lime, alkalis (efflorescence), pebbles, iron pyrites, organic matter.

Manufacturing: preparation of clay → moulding (hand/machine) → drying → burning (clamp ya kiln — Bull's trench, Hoffman's) at ~1000 °C. **Size:** standard modular brick 190×90×90 mm (nominal with mortar 200×100×100 mm); traditional 9" × 4½" × 3".

Class	Quality	Min. compressive strength	Water absorption (24 h)	Use
First class	Kiln burnt, uniform red, sharp edges, metallic ring	10.5 N/mm^2	≤ 20%	Exposed, load-bearing walls
Second class	Slightly irregular, hair cracks	7.0 N/mm^2	≤ 22%	Plastered walls
Third class	Under-burnt, soft	3.5 N/mm^2	≤ 25%	Temporary structures
Fourth class	Over-burnt, distorted	—	—	Broken as aggregate

Tests: compressive strength, water absorption, efflorescence (white salt patches — nil/slight acceptable), hardness (scratch by nail), soundness (do bricks takrane par clear ringing sound), shape & size (20 bricks ek line mein: 3800 ± 80 mm for modular), structure (broken face uniform, no voids).

3. Cement

OPC raw composition: lime CaO 60–67%, silica SiO_2 17–25%, alumina Al_2O_3 3–8%, iron oxide 0.5–6%, MgO 0.1–4%, SO_3 1–3%. **Gypsum** (~3%) setting **retarder** ke roop mein clinker grinding ke time add. **Bogue compounds:** C_3S (early strength, 7 days), C_2S (later strength), C_3A (flash set, max heat), C_4AF (little role). Manufacture: dry/wet process, rotary kiln ~1450 °C.

Types: OPC 33/43/53 grade (28-day strength in N/mm^2), Rapid hardening (high C_3S, early strength), Low heat (mass concrete — dams), Sulphate resisting (sea/sulphate soil), Portland Pozzolana Cement PPC (fly ash, durable), Portland slag cement, White cement (decoration), Quick setting (under-water), Hydrophobic, Coloured.

Test	Apparatus / method	IS requirement (OPC)
Fineness	90 µm IS sieve / Blaine air permeability	Residue ≤ 10%; specific surface ≥ 225 m^2/kg
Standard (normal) consistency	Vicat apparatus, 10 mm plunger penetrates to 5–7 mm from bottom	Usually 26–33% water
Initial / final setting time	Vicat needle (1 mm needle; annular collar for final)	Initial ≥ 30 min, Final ≤ 600 min
Soundness	Le Chatelier apparatus (autoclave for MgO)	Expansion ≤ 10 mm (autoclave ≤ 0.8%)
Compressive strength	70.6 mm mortar cubes (1:3 standard sand), 3/7/28 days	43 grade: 23, 33, 43 N/mm^2

Field tests: thanda feel, paani mein float nahi, no lumps, uniform grey colour. **Uses:** mortar, concrete, plaster, pipes, tiles.

4. Lime

Limestone ($CaCO_3$) ko jalane se **quick lime** CaO; paani milane (slaking) se **hydrated / slaked lime** $Ca(OH)_2$.

Hardening: CO_2 absorb karke wapas $CaCO_3$ (carbonation).

- **Fat (high calcium) lime:** CaO > 95%, slaking vigorous, volume 2–2.5 times, sets slowly in air only — whitewash, plaster.
- **Hydraulic lime:** clay 5–30%, paani mein bhi set — masonry mortar in damp places.
- **Poor (lean) lime:** impurities > 30%, poor quality — inferior mortar.

Tests: visual, heat test, acid test (hydraulicity), ball test, workability. Uses: mortar, plaster, whitewash, soil stabilisation, sand-lime bricks.

5. Timber

Types: Softwood (conifers — pine, deodar, chir; light, straight grain) vs **Hardwood** (teak, sal, sheesham, oak; strong, durable). Structure: pith, heartwood (dark, strong), sapwood, cambium, bark, medullary rays.

Properties of good timber: uniform colour, straight fibres, sweet smell, high strength, durability, no knots/defects, moisture content ~10–12% (seasoned), sonorous.

Seasoning (moisture kam karna): natural air seasoning, water seasoning, artificial — kiln, chemical, boiling, electrical. **Defects:** knots, shakes (heart/cup/star), rot (dry/wet), twist, warp, cup, bow. **Preservation:** creosote, ASCU, coal tar, paint. **Tests:** moisture content, compression/tension/bending strength. **Uses:** doors, windows, frames, formwork, furniture, plywood.

6. Concrete & Mortar

Concrete = cement + fine aggregate (sand, < 4.75 mm) + coarse aggregate (> 4.75 mm, usually 10–20 mm) + water (+ admixtures). **Mortar** = binder (cement/lime) + sand + water — bricks/stones jodne aur plaster ke liye. Cement mortar 1:3 to 1:6 (masonry), 1:4–1:6 (plaster); lime mortar 1:2 / 1:3.

Workability

Fresh concrete ki ease of mixing, placing, compacting aur finishing **without segregation**. Depends on water content, w/c ratio, aggregate size/shape/grading, admixtures.

- **Slump test:** frustum mould 300 mm high, 200 mm bottom, 100 mm top dia; 4 layers × 25 tamping; slump = height drop (25–75 mm normal RCC). Types: true, shear, collapse slump.
- **Compaction factor:** CF = wt. partially compacted / wt. fully compacted (0.75–0.95); low workability ke liye accurate.
- **Vee-Bee consistometer** (very stiff concrete, time in sec); **flow table** (high workability).

Strength Properties

- **Compressive strength:** 150 mm cubes, 28-day curing; characteristic strength f_{ck} (5% results neeche gir sakte). Grade **M20** = 20 N/mm^2.
- **Abram's law:** strength ∝ 1/(w/c ratio) — w/c badhao to strength ghatti. Usual w/c 0.4–0.6.
- **Tensile strength** ~ 10–15% of compressive; flexural strength $f_{cr} = 0.7\sqrt{f_{ck}}$ (IS 456).
- Other: durability, impermeability, shrinkage, creep, modulus $E_c = 5000\sqrt{f_{ck}}$.

Grade (nominal mix)	M5	M7.5	M10	M15	M20	M25
Cement : Sand : Coarse agg.	1 : 5 : 10	1 : 4 : 8	1 : 3 : 6	1 : 2 : 4	1 : 1.5 : 3	1 : 1 : 2

Nominal mix (fixed volume proportions) ordinary work ke liye; design mix (IS 10262) M20 aur upar important work ke liye. Dry volume ≈ 1.54 × wet volume. Use: M5–M10 PCC/levelling, M15 mild RCC, M20+ RCC (IS 456 min M20 for RCC).

> **PREPARATION OF CONCRETE** **Batching** (volume ya weight — weight batching accurate) → **Mixing** (hand mixing +10% extra cement; machine mixer ≥ 2 min) → **Transporting** (pan, wheel barrow, pump, transit mixer — segregation se bachao) → **Placing** (within initial setting, free fall ≤ 1.5 m, layers 150–300 mm) → **Compaction** → **Finishing** → **Curing**.

Compaction (entrapped air nikalna; 1% voids ~ 5–6% strength loss): hand rodding/tamping, **needle (immersion) vibrator** (most common), surface/screed vibrator (slabs), form (shutter) vibrator, table vibrator (precast).

Curing (cement hydration ke liye moisture + temperature maintain): ponding (slabs), wet gunny bags/hessian, sprinkling, curing compounds (membrane), steam curing (precast). Min **7 days** for OPC, **10 days** for PPC/blended (IS 456).

7. Elements of Building Construction

(a) Foundations

Foundation (substructure) structure ka load safely soil tak pahunchata hai, settlement uniform rakhta hai. **Shallow** (depth ≤ width): spread, combined, strap, raft. **Deep**: pile, well, caisson.

- **Conventional spread footing:** wall ya column ka base steps mein chauda karke load bade area par spread. Brick/stone masonry footing on lean concrete (PCC 1:4:8) bed; offset of each step ~ 50–60 mm. **Wall footing** (strip) aur **isolated column footing** (stepped/sloped).
- **RCC footings:** isolated (single column), **combined** (2 columns paas-paas ya property line ke paas), **strap** (do footings beam se jude), **raft/mat** (poora area slab — weak soil, heavy loads). Steel tension zone (bottom) mein.
- **Minimum depth (Rankine):** $h = (p/\gamma)\cdot[(1-\sin\varphi)/(1+\sin\varphi)]^2$; practically ≥ 0.9 m.

(b) Brick Masonry Walls

Bond = bricks ka arrangement jisse vertical joints continuous na ho. **Stretcher** = lambi face dikhe, **header** = chhoti face. Types: **English bond** (alternate courses of headers & stretchers; strongest, queen closer after first header), **Flemish bond** (har course mein header-stretcher alternate; better look; single/double Flemish), stretcher bond (half-brick partition walls), header bond (curved walls). Rules: bricks soak karke, joints ≤ 10 mm, lift ≤ 1.5 m/day, curing 7–10 days, plumb & level check.

(c) Plastering & Pointing

Plastering = wall/ceiling par mortar ki thin layer — smooth surface, protection from weather, appearance. Cement plaster 1:4 (external) / 1:6 (internal), thickness 12–20 mm (one/two coat). Types: lime, cement, mud, stucco, sand-faced, rough cast, pebble-dash. Defects: cracks, blistering, efflorescence, peeling.
Pointing = exposed masonry joints ko 10–20 mm rake karke rich mortar (1:2/1:3) se bharna — weather resistance + look. Types: flush, recessed, struck, V-grooved, weathered, tuck, beaded, rubbed.

(d) Floors

Floor ke parts: **sub-grade** (compacted soil), **base course** (PCC/brick bat), **flooring/topping**. Types: mud, brick, flagstone, cement concrete (IPS), terrazzo/mosaic, tiles (ceramic/vitrified), marble/granite/Kota stone, timber, PVC/linoleum. Selection: durability, cleanliness, cost, damp/sound/fire resistance, appearance.

(e) Roofs

Flat roofs (RCC slab, madras terrace, jack arch) — slope 1:100 drainage. **Pitched roofs:** lean-to/single (up to 2.5 m), couple (3.5 m), collar (4.5 m), **king post truss** (5–8 m), **queen post truss** (8–12 m), steel trusses (larger). **Shell** roofs (large spans). Covering: AC/GI sheets, tiles.

(f) Doors & Windows

Doors (min 2.0–2.1 m high, 0.9–1.0 m wide main): ledged, ledged-braced, ledged-braced-battened, framed & panelled, glazed, **flush** (most common), louvered, collapsible (steel, shops), rolling shutter (godowns), sliding, revolving (malls), swing.
Windows (glass area ~10–20% floor area): casement (side-hinged), double-hung/sash, pivoted, sliding, louvered, fixed, **bay** (projecting), **dormer** (sloping roof), clerestory, skylight, lantern, corner. Ventilator above for air.

(g) Lintels

Opening (door/window) ke upar horizontal member jo upar ki masonry ka load support karta hai. Types: timber, stone, brick (small span < 1 m), **RCC** (most common, precast/cast-in-situ), steel (I-section). Bearing = max(150 mm, lintel depth); depth ≈ span/12 (min 150 mm). Chhajja (weather shade) lintel ke saath cast. Arch = curved alternative.

(h) Staircases

Terms: tread (going), rise, nosing, flight (max 12–15 steps), landing, baluster, handrail, newel post, headroom (≥ 2.1 m). Good stair: **rise 150–190 mm**, **tread 250–300 mm**, **2R + T = 550–700 mm**, width 0.9–1.0 m (residential), 1.5–2 m (public), pitch 25°–40°.

Type	Description	Suitability
Straight	Ek direction mein, no turn	Narrow long hall, porch
Quarter-turn (L) / Half-turn	90° / 180° turn at landing	Residences
Dog-legged	Half-turn, flights adjacent, no gap	Limited space, houses (most common)
Open-newel (open well)	Half/quarter turn, well (gap) between flights	Public buildings, lift in well
Bifurcated	Wide flight splits into two	Large public buildings, theatres
Spiral / circular / geometrical	Winders around a post / curve	Very less space, emergency exit, architecture

UNIT V Surveying & Positioning

Surveying = earth surface par points ki relative position (horizontal distance, direction, elevation) determine karke map/plan banana. **Principles:** (1) **Whole to part** — pehle main control framework, phir details (errors accumulate nahi hote), (2) naya point kam se kam **do independent measurements** se locate karo. Classification: plane vs geodetic (earth curvature ignore / consider); instrument wise — chain, compass, plane table, theodolite, levelling, tacheometric, total station.

1. Surveying Instruments

(a) Chains & Tapes (linear measurement)

Chain	Length	Links	Remark
Metric chain	20 m / 30 m	100 / 150 (each 0.2 m)	Tallies every 5 m (20 m chain); brass handles
Gunter's chain	66 ft	100	Land measurement (10 sq chains = 1 acre)
Engineer's chain	100 ft	100	Each link 1 ft
Revenue chain	33 ft	16	Cadastral (revenue) surveys

Tapes: cloth/linen (stretch, rough work), metallic (cloth + brass wires, offsets), **steel** (accurate), **invar** (36% Ni-steel, very low thermal expansion — base lines, most accurate). **Accessories:** arrows (10 nos., mark chain lengths), ranging rods (2–3 m, red-white bands of 0.2 m), offset rods, pegs, plumb bob, line ranger, **cross staff / optical square** (right-angle offsets). **Ranging** = intermediate points ko line mein lana — direct (ends intervisible) / indirect (reciprocal, ends not visible). Chain survey: **triangulation** into well-conditioned triangles (angles 30°–120°); offsets (perpendicular/oblique). Errors: wrong chain length correction — $\text{True length} = (L'/L) \times \text{measured length}$ (L' = actual chain length, L = nominal).

(b) Levels

Horizontal line of sight dene wala instrument, elevations nikalne ke liye. Types: **dumpy level** (telescope fixed to vertical spindle — robust), tilting level, Wye (Y) level, **automatic (auto) level** (compensator), digital level. Parts: telescope, bubble tube, levelling head with foot screws, tripod. With **levelling staff** (self-reading — telescopic/folding 4 m; target staff). **Temporary adjustments:** (1) setting up, (2) levelling up (bubble centre — two foot screws parallel, phir third), (3) focusing eyepiece & object (elimination of **parallax**).

(c) Theodolite

Sabse precise angle-measuring instrument — **horizontal & vertical angles**, also prolonging lines, levelling, traversing. Parts: telescope, vertical circle, horizontal (upper & lower) plates, vernier (least count usually 20″), upper/lower clamps & tangent screws, plate levels, tribrach, trivet, plumb bob. **Transit** theodolite: telescope vertical plane mein 180° ghoom sakta hai. Terms: face left/face right, swinging, transiting, line of collimation. Temporary adjustments: setting up, centering, levelling, focusing. **Methods:** repetition (accurate single angle, vernier error kam), reiteration (several angles at a station). Face left + face right ka average instrumental errors cancel karta hai.

(d) Plane Table & Related Devices

Plane table survey: field mein hi plotting (graphical) — small scale maps, quick. Accessories: drawing board on tripod, **alidade** (plain/telescopic, sighting), spirit level, trough compass (orientation), plumbing fork + plumb bob (centering), drawing sheet. Setting: levelling, centering, **orientation** (by trough compass or back-sighting — most accurate). **Methods:** radiation (ek station se rays + distance), intersection (do stations se rays ka intersection — inaccessible points), traversing, **resection** (table ki khud ki position — two-point / three-point problem). Pros: no field book, fast; cons: wet weather mein mushkil, less accurate.
Related modern devices: EDM (electronic distance measurement), **Total Station** (electronic theodolite + EDM + microprocessor/data storage), GPS/GNSS (satellite positioning), digital levels, drones/LiDAR.

2. Measurement of Directions — Bearings & Compass

Bearing = kisi line ka horizontal angle ek reference (meridian) direction se. Meridians: **true** (geographical N-S, fixed), **magnetic** (compass needle direction), arbitrary. **Magnetic declination** = true & magnetic meridian ke beech angle: $\text{True bearing} = \text{Magnetic bearing} \pm \text{declination}$ (+ if E, − if W). **Dip** = needle ka horizontal se inclination.

- **WCB (Whole Circle Bearing):** North se clockwise 0°–360° (prismatic compass).
- **QB / Reduced Bearing:** N ya S se E ya W ki taraf 0°–90°, e.g. N 30° E (surveyor's compass).
- **Fore bearing (FB)** = line ki direction of progress mein; **Back bearing BB = FB ± 180°** (FB < 180° → +, > 180° → −). QB mein: N↔S aur E↔W swap, angle same.
- **Included angle** = difference of bearings of two lines meeting at a point.

WCB range	Quadrant	QB
0°–90°	NE	N θ E
90°–180°	SE	S (180° − θ) E
180°–270°	SW	S (θ − 180°) W
270°–360°	NW	N (360° − θ) W

Example: WCB 135° → S 45° E; 210° → S 30° W; 290° → N 70° W. FB of AB = 52° → BB = 232°.

Prismatic vs Surveyor's compass: prismatic mein graduated ring needle ke saath ghoomta hai, WCB, 0° at S end (inverted figures), prism se sighting + reading ek saath, tripod optional; surveyor's mein ring box ke saath fixed, needle ghoomti hai, QB, 0° at N & S (E-W interchanged), sighting aur reading alag, tripod essential.

Local attraction: nearby iron/electric cables needle ko deflect karte hain. Detection: kisi line ka FB − BB ≠ 180°. Correction: jis line ka FB − BB = 180° exactly ho, uske stations free maan kar baaki bearings correct karo (ya included-angle method). **Other methods of measuring directions:** theodolite se included angles / deflection angles (traverse), azimuth by astronomical observation, total station / GPS.

3. Measurement of Elevations — Levelling

TERMS **Datum** = reference surface (MSL); **Reduced Level (RL)** = datum se point ki height; **Bench Mark (BM)** = known RL wala fixed point (GTS, permanent, arbitrary, temporary); **Back Sight (BS)** = known RL point par pehli reading after setting instrument; **Fore Sight (FS)** = instrument shift karne se pehle last reading; **Intermediate Sight (IS)** = beech ki readings; **Change Point (CP)** = jahan FS aur BS dono li jaati hain; **Height of Instrument (HI)** = line of collimation ka RL.

HI (COLLIMATION) METHOD HI = RL of BM + BS; RL = HI − (IS or FS)
Check: ΣBS − ΣFS = Last RL − First RL
Fast; IS ka check nahi — profile levelling ke liye.

RISE & FALL METHOD Prev. reading − current reading: +ve = Rise, −ve = Fall; RL = prev. RL + rise − fall
Check: ΣBS − ΣFS = ΣRise − ΣFall = Last RL − First RL
IS ka bhi check — accurate.

Example: BM RL = 100.000 m. Readings: 1.500 (BS on BM), 2.300 (IS), 1.800 (FS) & 1.200 (BS) at CP, 0.650 (FS).

Point	BS	IS	FS	HI	Rise	Fall	RL
BM	1.500			101.500			100.000
P		2.300				0.800	99.200
CP	1.200		1.800	100.900	0.500		99.700
Q			0.650		0.550		100.250
Σ	**2.700**		**2.450**		**1.050**	**0.800**	**Check: 0.250 = 0.250 = 0.250 ✓**

4. Reciprocal Levelling

Jab do points A, B ke beech **river, valley ya pond** ho (beech mein instrument nahi lag sakta), tab reciprocal levelling. Instrument pehle A ke paas lagao → readings a_1 (A par), b_1 (B par); phir B ke paas lagao → a_2, b_2.

FORMULA

$$\text{True difference } H = [(b_1 - a_1) + (b_2 - a_2)] / 2 \quad (\text{B, A se H neeche})$$

$$\text{Combined error } e = [(b_1 - a_1) - (b_2 - a_2)] / 2$$

Ye method **collimation error, curvature aur refraction** (average) errors ko eliminate karta hai.

Example: a_1 = 1.625, b_1 = 2.545, a_2 = 0.725, b_2 = 1.405; RL_A = 100.000. H = (0.920 + 0.680)/2 = **0.800 m** → RL_B = **99.200 m**; error e = (0.920 − 0.680)/2 = 0.120 m.

Other levelling types: simple, differential/fly (door ke points, change points ke saath), profile (L-section) & cross-section, check levelling, trigonometric (vertical angles se), barometric. Curvature correction $C_c = -0.0785\ d^2$, refraction $C_r = +0.0112\ d^2$, combined = $-0.0673\ d^2$ (d in km, result in m).

EXAM TIP Levelling table (HI + rise-fall, arithmetic check), WCB ↔ QB conversion aur compass comparison almost har saal aate hain.